

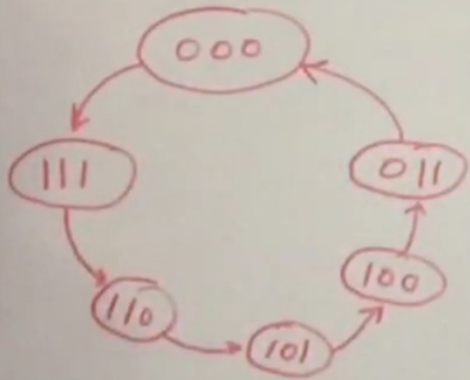
MOD-6 DOWN SYNCHRONOUS COUNTER

Step 1 :- MOD-6

6 States

3 Bit = 3 FF, JK FF

Step 3: State Diagram



Step 2: Ex. JK FF

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

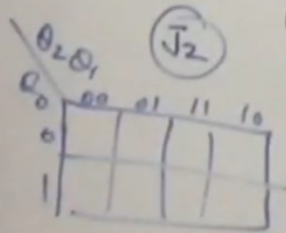
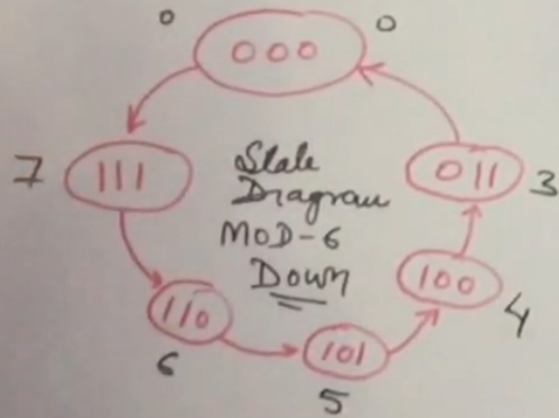
MOD-6 DOWN SYNCHRONOUS COUNTER

Step 1 :- MOD-6

⑥ States

3 bit = 3 FF, JK FF

Step 3: State Diagram



Step 2: Ex. JK FF

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

P.S.	N.S.	I/P			
		J ₂ k ₂	J ₁ k ₁	J ₀ k ₀	
0 0 0	1 1 1	1 X	1 X	1 X	
1 1 1	1 1 0	X 0	X 0	X 1	
1 1 0	1 0 1	X 0	X 1	1 X	
1 0 1	1 0 0	X 0	0 X	X 1	
1 0 0	0 1 1	X 1	1 X	1 X	
0 1 1	0 0 0	0 X	X 1	X 1	

MOD-6 DOWN SYNCHRONOUS COUNTER

Step 1 :- MOD-6

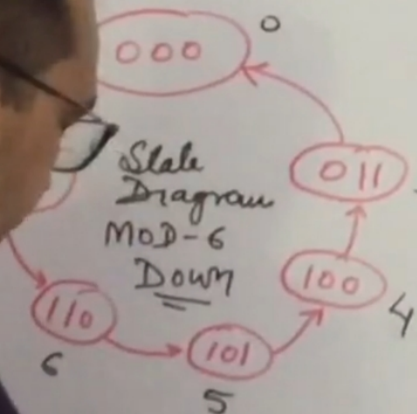
6 States

3 Bit = 3 FF, JK FF

Step 2: Ex. JK FF

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

3: State Diagram



	PS $Q_2 Q_1 Q_0$	NS $Q_2 Q_1 Q_0$	I/P		
			$J_2 K_2$	$J_1 K_1$	$J_0 K_0$
0	0 0 0	1 1 1	1 X	1 X	1 X
1	1 1 1	1 1 0	X 0	X 0	X 1
2	1 1 0	1 0 1	X 0	X 1	1 X
3	1 0 1	1 0 0	X 0	0 X	X 1
4	1 0 0	0 1 1	X 1	1 X	1 X
5	0 1 1	0 0 0	0 X	X 1	X 1

J_2

$Q_2 Q_1$	00	01	11	10
Q_0	1	X	X	X
	1	X	0	X

$$K_2 = \bar{Q}_1 \cdot \bar{Q}_0$$

$$J_2 = \bar{Q}_0$$

J_1

$Q_2 Q_1$	00	01	11	10
Q_0	1	X	X	1
	1	X	X	0

$$J_1 = \bar{Q}_0$$

K_1

$Q_2 Q_1$	00	01	11	10
Q_0	1	X	X	1
	1	X	1	0

$$K_1 = \bar{Q}_2 + \bar{Q}_0$$

$$J_0 = K_0 = 1$$

$$J_1 = \bar{Q}_0$$

$$K_1 = \bar{Q}_2 + \bar{Q}_0$$

$$J_2 = \bar{Q}_0$$

MOD-6 DOWN SYNCHRONOUS COUNTER

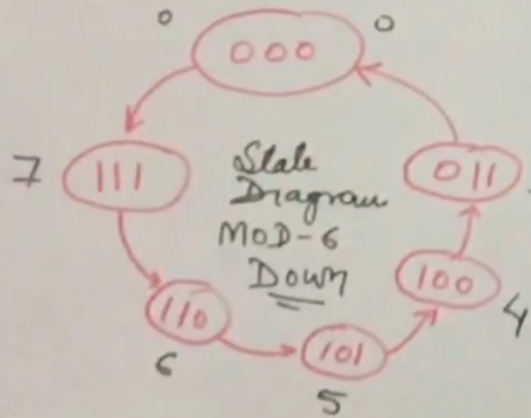
Step 1 :- MOD-6

6 States

3 bit = 3 FF, JK FF

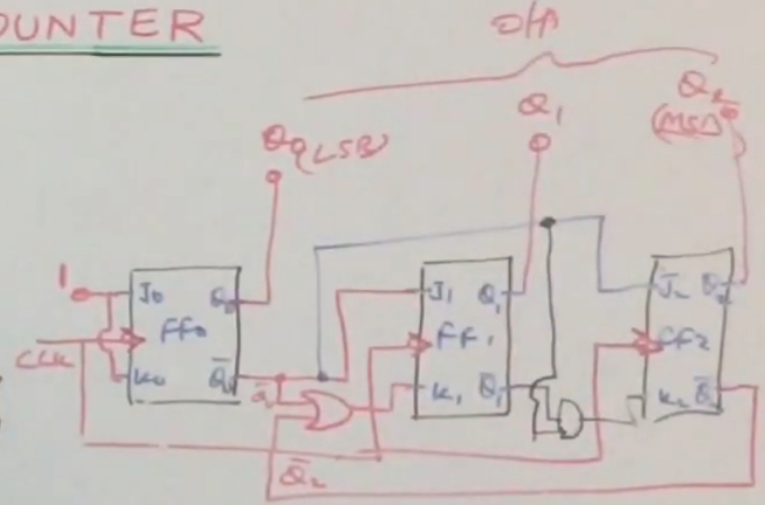
Step 2: Ex JK FF

Step 3: State Diagram



Outputs

	PS Q_2, Q_1, Q_0	NS Q_2, Q_1, Q_0
0	0 0 0	1 1 1
1	1 1 1	1 1 1
2	1 1 0	1 1 1
3	1 0 1	1 1 1
4	1 0 1	1 1 1
5	1 0 1	1 1 1
6	0 1 1	1 1 1



Truth Table for J_2 :

Q_2, Q_1	00	01	11	10
0	1	x	x	x
1	x	0	x	x

$K_2 = \overline{Q_1} \cdot \overline{Q_0}$

Truth Table for J_2 :

Q_2, Q_1	00	01	11	10
0	x	x	0	1
1	x	x	0	0

$K_0 = 1$